



# BICOL UNIVERSITY
# COLLEGE OF ENGINEERING

## DEPARTMENT OF CHEMICAL ENGINEERING
## CHEM 13 - ORGANIC CHEMISTRY
## ASYNCHRONOUS ACTIVITY 2


**ANDREY S. RICAHUERTA**

**BS CHEMICAL ENGINEERING 2A**


$F = \frac{Gm_1m_2}{r^2}$

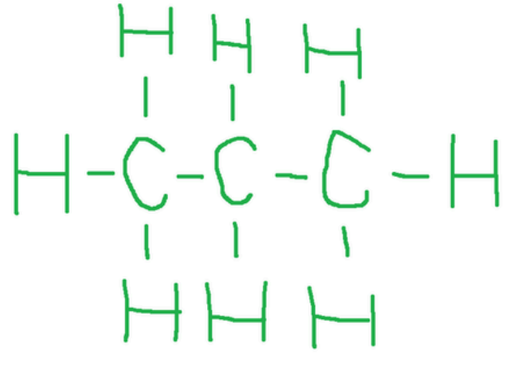
$4H_2 \times \frac{2H_2O}{2H_2} =$

# CARD 01 - Propane
# Molecular Formula: $C_3H_8$

## 1. Molecular Formula

$$C_3H_8$$

## 2. Lewis Structure



## 3. Condensed Structural Formula

$$CH_3CH_2CH_3$$

## 4. Bond-Line Formula (Ketcher)

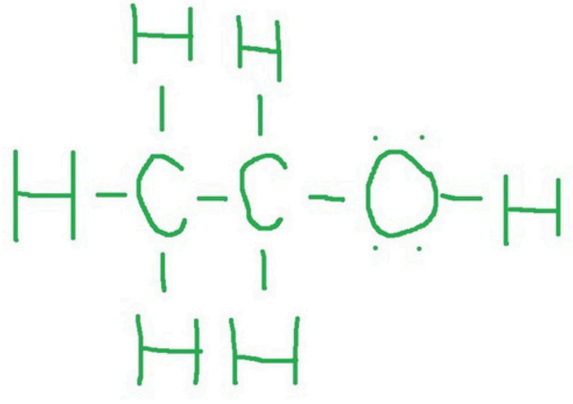


# CARD 02 - Ethanol
# Molecular Formula: $C_2H_6O$

## 1. Molecular Formula

$$C_2H_6O$$

## 2. Lewis Structure



## 3. Condensed Structural Formula

$$CH_3CH_2OH$$

## 4. Bond-Line Formula (Ketcher)

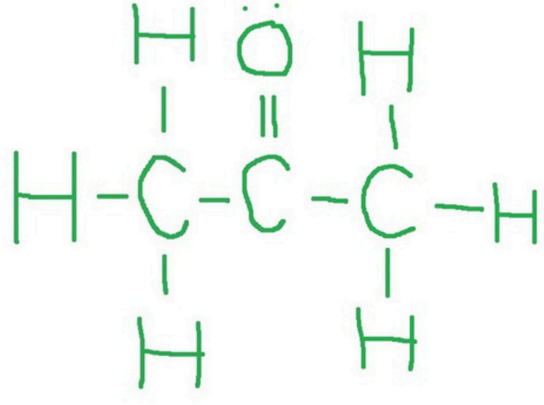


# CARD 03 - Acetone (propan-2-one)
# Molecular Formula: $C_3H_8$

## 1. Molecular Formula

$$C_3H_6O$$

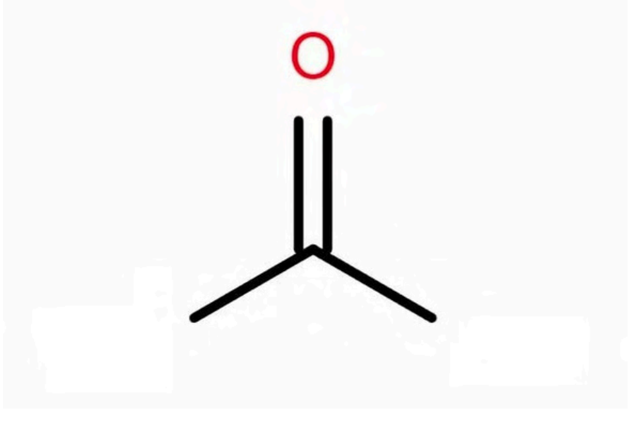
## 2. Lewis Structure



## 3. Condensed Structural Formula

$$CH_3COCH_3$$

## 4. Bond-Line Formula (Ketcher)

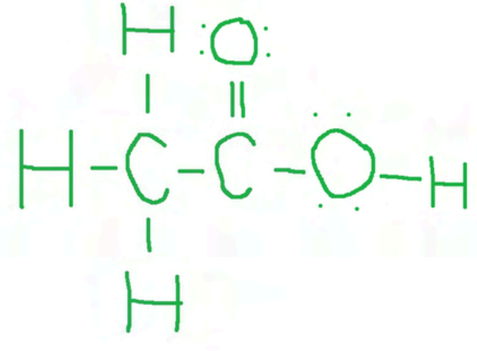


# CARD 04 - Acetic acid (ethanoic acid)
# Molecular Formula: $C_2H_4O_2$

## 1. Molecular Formula

$$C_2H_4O_2$$

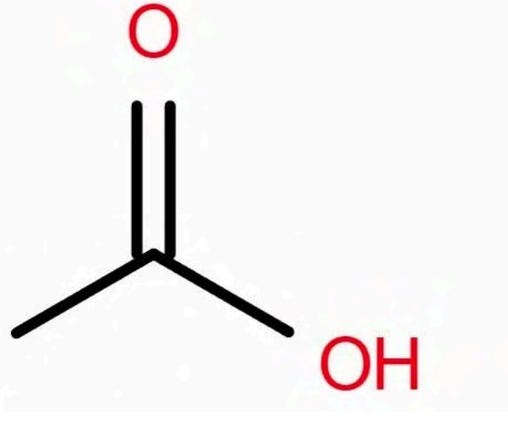
## 2. Lewis Structure



## 3. Condensed Structural Formula

$$CH_3COOH$$

## 4. Bond-Line Formula (Ketcher)

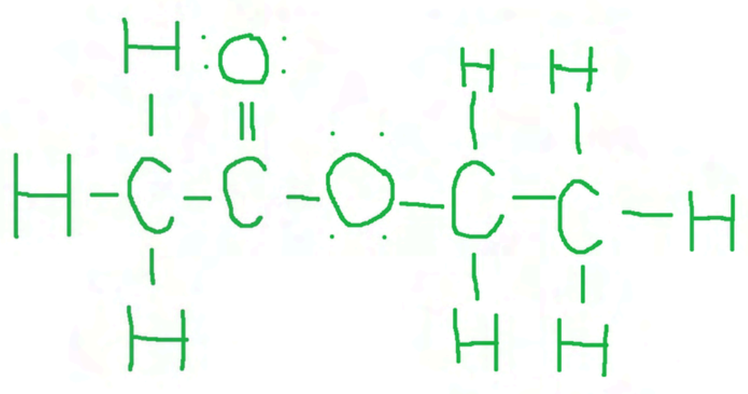


# CARD 05 - Ethyl Acetate (ethyl ethanoate)
# Molecular Formula: $C_4H_8O_2$

## 1. Molecular Formula

$$C_4H_8O_2$$

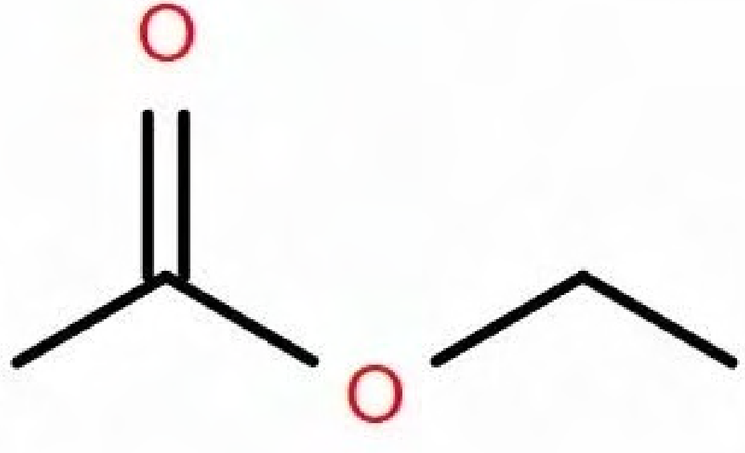
## 2. Lewis Structure



## 3. Condensed Structural Formula

$$CH_3COOCH_2CH_3$$

## 4. Bond-Line Formula (Ketcher)

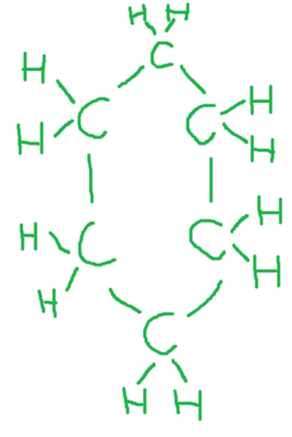


# CARD 06 - Cyclohexane
# Molecular Formula: $C_6H_{12}$

## 1. Molecular Formula

$$C_6H_{12}$$

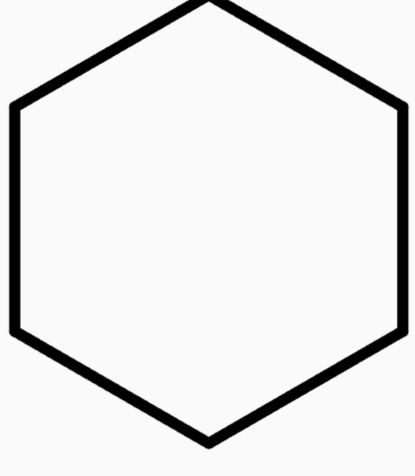
## 2. Lewis Structure



## 3. Condensed Structural Formula

$$(CH_2)_6$$

## 4. Bond-Line Formula (Ketcher)

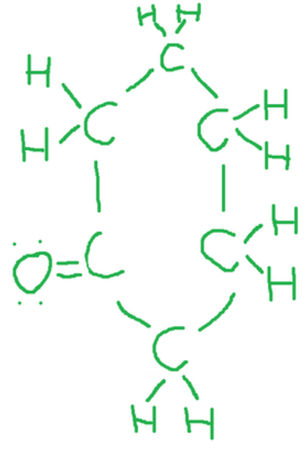


# CARD 07 - Cyclohexanone
# Molecular Formula: $C_6H_{10}O$

## 1. Molecular Formula

$$C_6H_{10}O$$

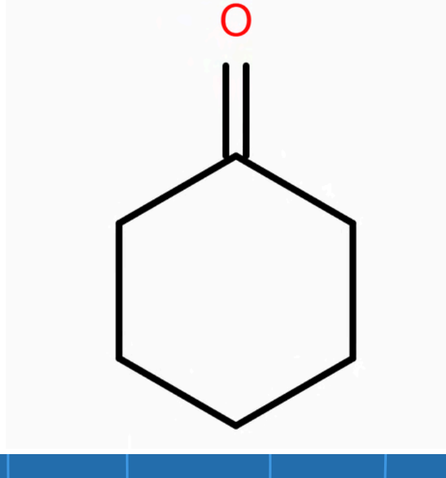
## 2. Lewis Structure



## 3. Condensed Structural Formula

$$(CH_2)_5CO$$

## 4. Bond-Line Formula (Ketcher)

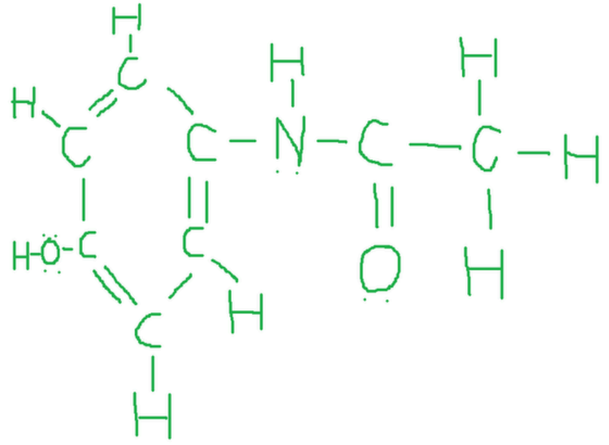


# CARD 08 - Acetaminophen (paracetamol)
# Molecular Formula: $C_5H_9NO_2$

## 1. Molecular Formula

$$C_5H_9NO_2$$

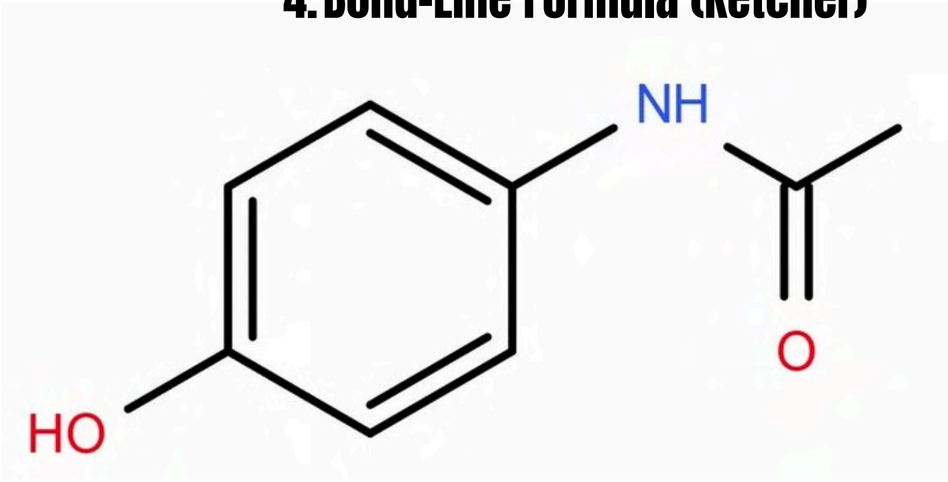
## 2. Lewis Structure



## 3. Condensed Structural Formula

$$HOC_6H_4NHCOCH_3$$

## 4. Bond-Line Formula (Ketcher)

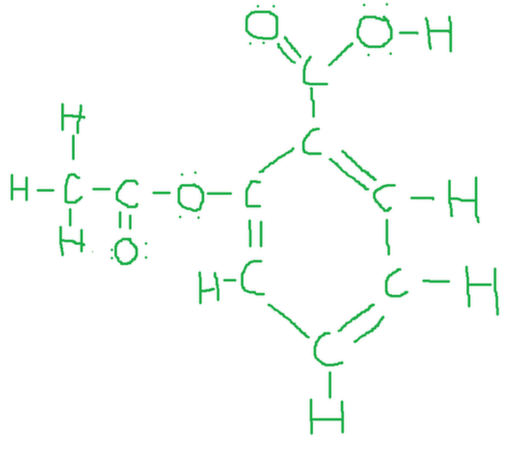


# CARD 09 - Aspirin (acetylsalicylic acid)
# Molecular Formula: $C_9H_8O_4$

## 1. Molecular Formula

$$C_9H_8O_4$$

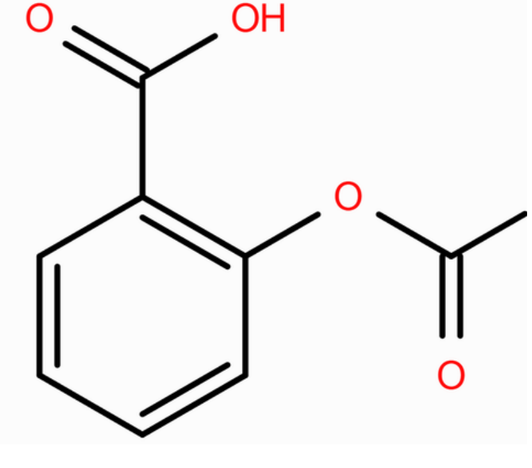
## 2. Lewis Structure



## 3. Condensed Structural Formula

$$CH_3COOC_6H_4COOH$$

## 4. Bond-Line Formula (Ketcher)



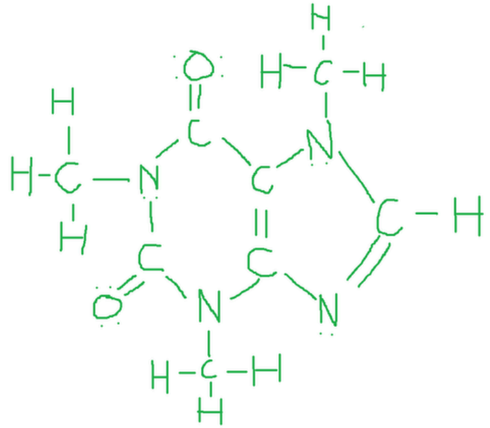
# CARD 10 - Caffeine

## Molecular Formula: $C_8H_{10}N_4O_2$

### 1. Molecular Formula

$$C_8H_{10}N_4O_2$$

### 2. Lewis Structure



### 3. Condensed Structural Formula

$$(CH_3)_3C_5HN_4O_2$$

### 4. Bond-Line Formula (Ketcher)

